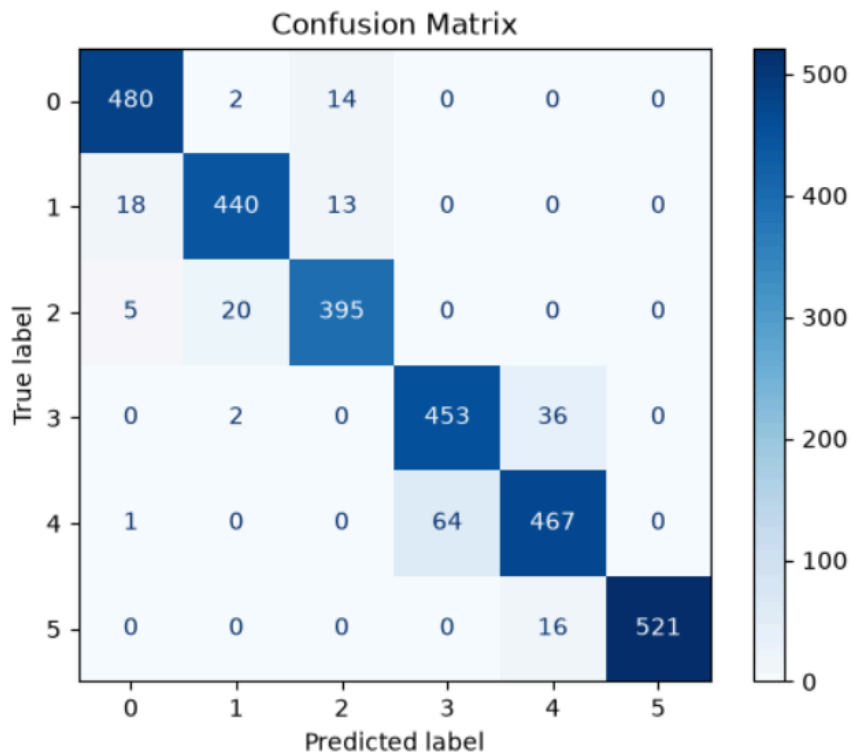
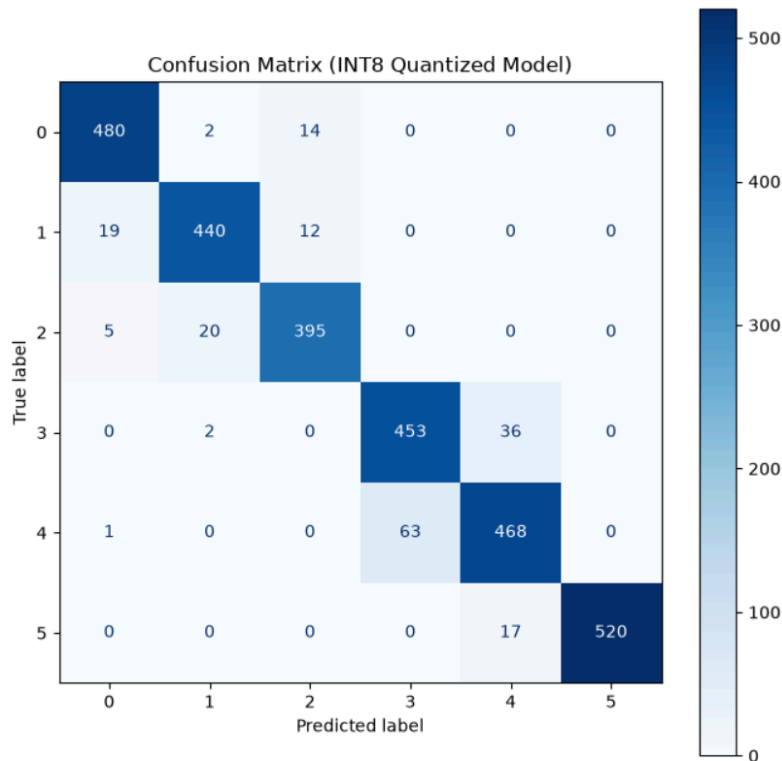


Quantization:

1. I got a similar model size from dynamic quantization and from int8 quantization.
2. All quantization methods yielded similar accuracy within my testing datasets.
3. No, the models both had similar size and accuracy. I would not say there was an improvement.
4. It is a compact dataset and trains fast. It did not lose much accuracy when quantizing the dataset.
5. I would choose the Int8 (PTQ) quantization. It was the smallest model and retained the same amount of accuracy as the others models. I chose it over the dynamic because it should have faster execution speed and I chose it over the QAT to reduce complexity.

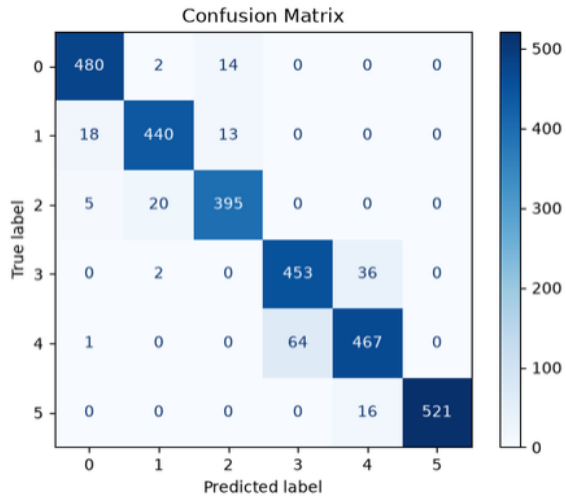




	Model Family	Format	Model Size (KB)	Test Accuracy
0	Baseline DNN	FP32	726.664062	0.935188
1	Baseline DNN	DYNAMIC	186.062500	0.935867
2	Baseline DNN	FLOAT16	365.625000	0.935188
3	Baseline DNN	INT8	185.281250	0.935188
4	Baseline DNN	QAT INT8	185.632812	0.933831

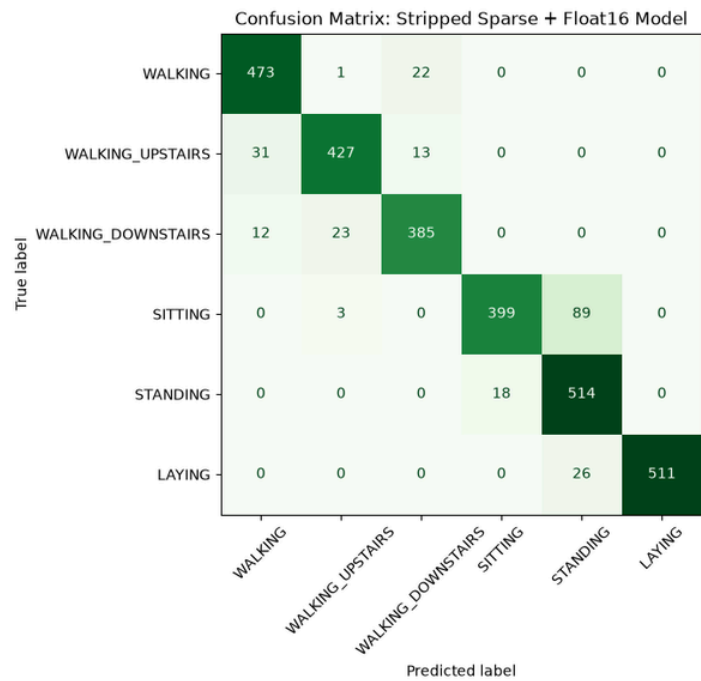
Pruning with quantization:

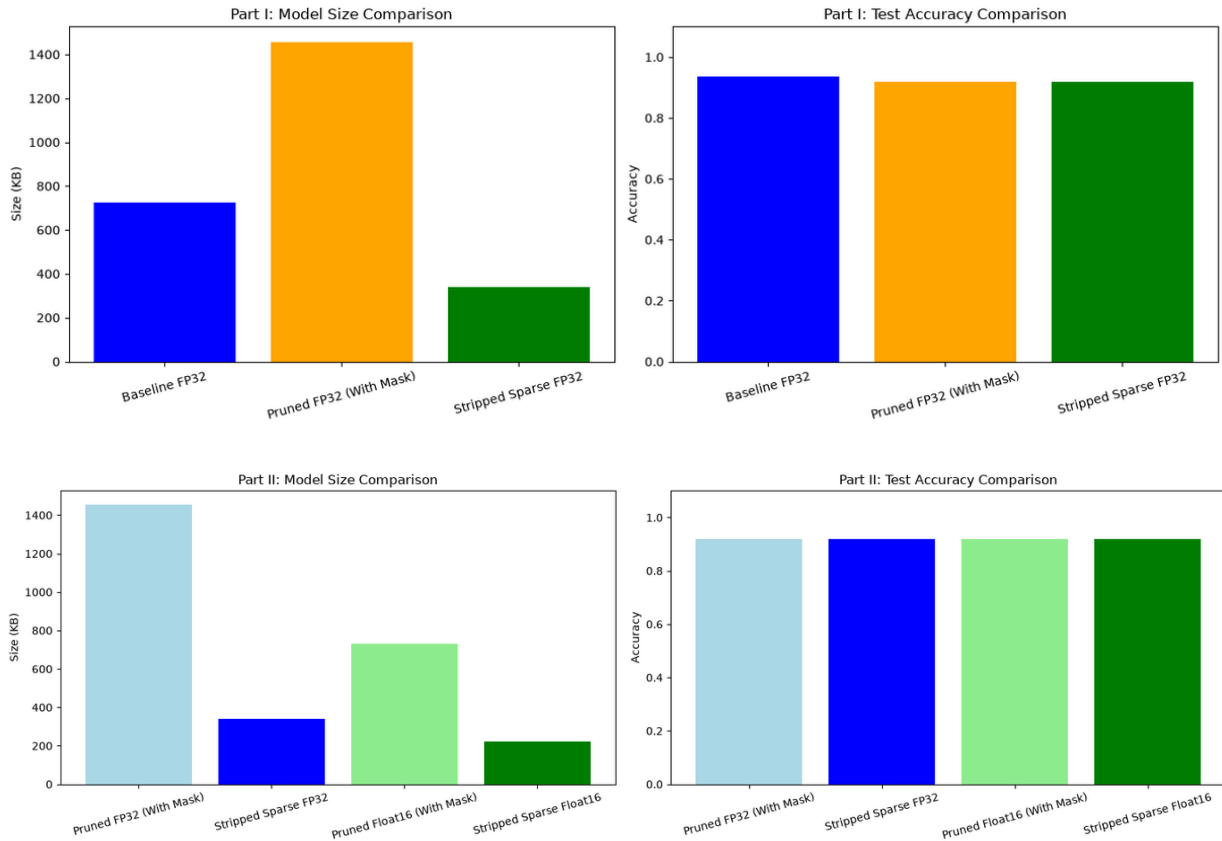
1. No, pruning alone actually made the file size larger than the baseline size was. The pruned model still contained wrapper information, so the model contained the original weights and the extra masking variables.
2. Stripping the pruning wrappers removes the masking that is used for training the model. Stripping creates an actual sparse model and it is necessary to actually reduce the size of the model and compress the redundant connections.
3. The Stripped Sparse Float16 model had the smallest file size at around 223 KB.
4. No, the accuracy stayed the same. Float16 quantization preserves accuracy very well compared to Int8 quantization.
5. I would choose the Stripped Sparse Float16 model because it has the smallest size and maintains high accuracy.



Classification Report (Stripped Sparse Float16):

	precision	recall	f1-score	support
WALKING	0.9167	0.9536	0.9348	496
WALKING_UPSTAIRS	0.9405	0.9066	0.9232	471
WALKING_DOWNSTAIRS	0.9167	0.9167	0.9167	420
SITTING	0.9568	0.8126	0.8789	491
STANDING	0.8172	0.9662	0.8854	532
LAYING	1.0000	0.9516	0.9752	537
accuracy			0.9192	2947
macro avg	0.9246	0.9179	0.9190	2947
weighted avg	0.9244	0.9192	0.9195	2947



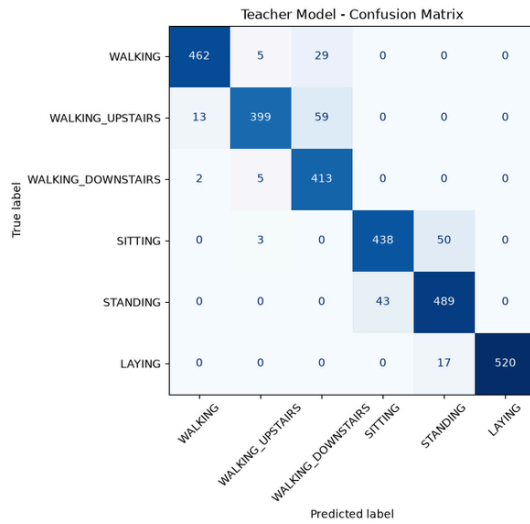


Distillation:

1. The distilled student outperformed the baseline student. It had a higher accuracy of 93.62% compared to the baseline of 92.50% accuracy.
2. Yes, the knowledge distillation was effective. The model reached a higher accuracy than the teacher model and the baseline, while being a smaller size.
3. The model size was much smaller after pruning and quantization. It shrunk from 151KB to only 37KB after pruning, and then to 21KB after quantization. However, the accuracy dropped to around 81%.
4. I would choose the stripped sparse distilled TFLite model. At only 21KB it is the smallest model while still maintaining good accuracy. If the device had memory to spare, I would choose the distilled model, because of its high accuracy, but it takes 151KB of memory.
5. The final INT8 model is a good TinyML candidate because of the extremely small size after using three different compression techniques: distillation, pruning, and quantization. The small model size and good accuracy is perfect for deployment on a low power edge computing device.

Teacher Test Accuracy: 0.9233

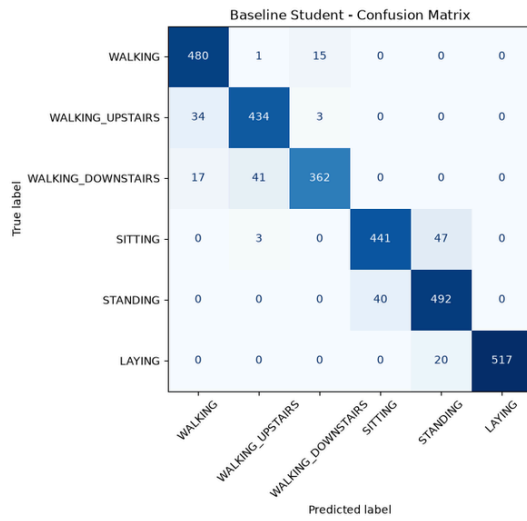
	precision	recall	f1-score	support
WALKING	0.9686	0.9315	0.9496	496
WALKING_UPSTAIRS	0.9684	0.8471	0.9037	471
WALKING_DOWNSTAIRS	0.8244	0.9833	0.8969	420
SITTING	0.9186	0.8921	0.9012	491
STANDING	0.8795	0.9192	0.8989	532
LAYING	1.0000	0.9683	0.9839	537
accuracy			0.9233	2947
macro avg	0.9252	0.9236	0.9224	2947
weighted avg	0.9288	0.9233	0.9238	2947



Test Accuracy: 0.9250

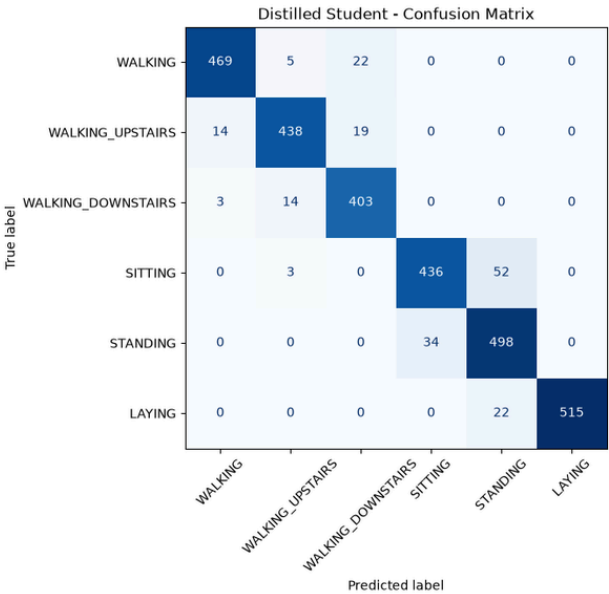
Classification Report:

	precision	recall	f1-score	support
WALKING	0.9848	0.9677	0.9348	496
WALKING_UPSTAIRS	0.9861	0.9214	0.9137	471
WALKING_DOWNSTAIRS	0.9526	0.8619	0.9058	420
SITTING	0.9168	0.8982	0.9074	491
STANDING	0.8881	0.9248	0.9019	532
LAYING	1.0000	0.9628	0.9810	537
accuracy			0.9250	2947
macro avg	0.9266	0.9228	0.9240	2947
weighted avg	0.9266	0.9258	0.9251	2947



Distilled Student Test Accuracy: 0.9362

	precision	recall	f1-score	support
WALKING	0.9650	0.9456	0.9552	496
WALKING_UPSTAIRS	0.9522	0.9299	0.9409	471
WALKING_DOWNSTAIRS	0.9077	0.9595	0.9329	420
SITTING	0.9277	0.8880	0.9074	491
STANDING	0.8706	0.9361	0.9022	532
LAYING	1.0000	0.9590	0.9791	537
accuracy			0.9362	2947
macro avg	0.9372	0.9364	0.9363	2947
weighted avg	0.9379	0.9362	0.9365	2947



	Model	Format	Test Accuracy	Model Size (KB)
0	Distilled Student TFLite	FP32	0.936206	151.476562
1	Pruned Distilled TFLite (with mask)	FP32	0.807940	304.253906
2	Stripped Sparse Distilled TFLite	FP32 + Sparse	0.807940	36.894531
3	Stripped Sparse Distilled TFLite	INT8 + Sparse	0.807940	20.625000

Final Distilled + Pruned + INT8 Model - Confusion Matrix

